

Erdős Problem #978

EP-978

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $f \in \mathbb{Z}[x]$ be an irreducible polynomial of degree $k > 2$ (and suppose that $k \neq 2^l$ for any $l \geq 1$) such that the leading coefficient of f is positive.

Does the set of integers $n \geq 1$ for which $f(n)$ is $(k-1)$ -power-free have positive density?

Are there infinitely many n for which $f(n)$ is $(k-2)$ -power-free?

In particular, does $n^4 + 2$ represent infinitely many squarefree numbers?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>